Korash Assani

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Education

University of Cincinnati — **Cincinnati, OH**
B.S. in Astrophysics — *August 2016 – May 2020*
B.S. in Physics — GPA: 3.769/4.000
B.A. in Mathematics

University of Virginia — **Charlottesville, VA**
Ph.D. in Astronomy — *August 2020 – May 2026*

Research Experience

Postdocoral Researcher — **Bern, Switzerland**
Advisor: Maria N. Drazdovskaya — *Sep. 2026-Current*

Graduate Research Assistant — **Charlottesville, VA**
Advisor: Zhi-Yun Li — *June, 2020-Aug. 2026*

Undergraduate Research Assistant — **Cincinnati, OH**
Advisor: Michael L. Sitko — *August, 2016-May, 2020*

Invited Talks

- **Stellar Fountains: Probing Star Formation and Dust with JWST Observations of Protostellar Outflows** Sep, 2025
  University of Cincinnati Astronomy Seminar Series Cincinnati, OH
- **Protostellar Outflows with JWST: From Line Diagnostics to Dust Attenuation** Sep, 2025
  Space Telescope Science Institute - Dust/ISM Group Baltimore, ML
- **The Birth of Stars: JWST Insights into Protostellar Outflows and Dust in Star-Forming Regions** Mar, 2025
  Virginia Tech Astronomy Series Blacksburg, VA

Contributed Presentations

- **Ro-vibrational H_2 as a Probe of Cosmic Rays in Protoplanetary Disks** Oct. 2026
  Palazzo della Carovana Pisa, Italy
- **[Fe II] & H_2 Excitation Conditions of the TMC1A Protostellar Outflow** Sep, 2024
  Specola Vaticana Castel Gandolfo, Italy
- **Global Simulations of Planetary Growth via Pebble Accretion**
  - Gordon Research Conference, June 2023 Mount Holyoke, MA
  - Bob Rood Symposium, April 2023 Charlottesville, VA
  - VICO-CICO Workshop, Nov 2021 Charlottesville, VA
  - Sagan Exoplanet Summer Virtual Workshop, Poster #54, June 2021
- **Variability in the Gas and Dust Emission of the UX Orionis Star CQ Tau** Jan, 2020
  235th American Astronomical Society Meeting, Poster #451.01 Honolulu, HI
- **Modeling the Circumstellar Disk of HD 166191**

- 233rd American Astronomical Society Meeting, Poster #163.19, Jan 2019 — Seattle, WA
- UC Department of Physics, MUSE Fellowship Presentation, Nov 2018 — Cincinnati, OH

Workshops

- **NASA GPU Hackathon** — Sep 12, 20-28, 2022
  Remote Workshop
- **Dynamic and Chemical Connection Workshop** — July 4-8, 2022
  Lorentz Center — Leiden, Netherlands

Observing Proposals

- **The Dark Side of the Force: Unraveling Protostellar Jet Asymmetry by Probing TMC1A's Fainter Red-shifted Outflow with JWST.** Assani, Korash D., Zhi-Yun Li, Daniel Harsono, et al., Proposal ID: 8872 | Accepted, JWST Cycle 4 (2025), 7.5 hrs, up to ~$100,000, See Universe Today highlight
- **Is the Abnormally Low Spectral Index of the Elias 2-27 Disk Caused by Dust Scattering?** Assani, Korash D., Daniel Lin, et al., Project Code: 2023.1.00377.S | ALMA Cycle 10 (2023), Accepted Rank-C, Not Observed
- **ALMA Meets JWST: Is There Warm Molecular Gas Near the [Fe] Jet?** Harsono, Daniel, Korash Assani, et al., Project Code: 2024.1.00046.S | Accepted, Rank C
- **Imaging Planet Formation at Its Earliest Stages: Measuring the Extinction Level of an Enshrouded Protoplanet** Wagner, Kevin, Korash Assani, et al., Proposal ID: 4010 | Accepted, JWST Cycle 2 (2022)

Teaching Experience

News Highlight: UC Triple Major Seeks to Inspire and Educate

- **UVA Center for Teaching Excellence**
  - Tomorrows Professor Today (TPT), Jan 2025 – Present
  - Teaching as a Graduate Student (TAGS), Aug 2021
- **Teaching Assistant** — Fall 2020, Spring 2022
  University of Virginia, Astronomy Department
- **Learning Commons Instructor** — Jan 2017 – May 2019
  University of Cincinnati, Learning Commons Roles:
  - **Peer Leader** (Aug 2018 – May 2019)
  - **Peer Tutor** (Aug 2018 – May 2019)
  - **Supplemental Instructor** (Aug 2017 – Apr 2018)
  - **Learning Assistant** (Jan 2017 – Apr 2017)
- **Teaching Assistant** — Aug 2017 – Dec 2017
  University of Cincinnati, Physics Department

Leadership Roles

- **UVA Astro Grad Lunch** — Jan 2023 – Jan 2024
  Role: Coordinator
- **UVA Astronomy Graduate Journal Club** — Jan 2021 – Jan 2023

Role: Coordinator

- **Society of Physics Students** Aug 2018 – May 2020
  Role: Vice President
- **UC College of Arts and Sciences Student Ambassadors** Dec 2016 – May 2020
  Roles: Treasurer, Vice President, President
- **Circle K International** Aug 2016 – Apr 2018
  Role: Head of Recruitment

Volunteer and Outreach

- **Virginia Piedmont Regional Science Fair** Mar 30, 2023, Mar 20, 2025
  Judge
- **Public Nights at McCormick Observatory** 2021 – Present
  Telescope Operator
- **Dark Sky Bright Kids** 2021, 2022
  Star Party Volunteer, Semester Club
- **University of Virginia, Astronomy Mentoring Program** Aug 2021 – Aug 2022
  Undergraduate Mentor
- **Astronomy on Tap** Sep 12, 2022
  Speaker: *"JWST: Exploring the Universe Like Never Before"*
- **Cincinnati Observatory** Nov 2016 – Aug 2017
  Volunteer Docent

Academic Honors

- **Sigma Pi Sigma**, Physics Honor Society 2019
- **Phi Beta Kappa**, National Honor Society 2020
- **Magna Cum Laude**, University of Cincinnati 2020
- **Distinguished University Honors Scholar**, University of Cincinnati 2020

Computational Experience

- Extensive experience in **Python**, **Fortran**, and **Mathematica**, with proficiency in **IDL**, **C++**, **MATLAB**, **HTML**, and **Java**. Comfortable adapting to new programming languages as needed.
- Experienced in computationally intensive **3D physics simulations**, including: - *Hydrodynamic + dust simulations* of planet formation (**DISPATCH**, **Athena++**). - *Atomic spectral synthesis modeling* of [Fe II] emission (**CLOUDY**). - *Monte Carlo radiative transfer modeling* of full spectral energy distributions (**HOCHUNK3D**).
- Proficient in **Python**-based data analysis and visualization of large datasets, including **N-dimensional datacubes** from **JWST** and **ALMA** observations.
- Experienced in **Git** version control using GitHub and Bitbucket: `https://github.com/KorashAssani`

Publications

Peer-Reviewed Publications – First Author

2026: **JWST Edge-on Disk Ice (JEDIce): Vibrationally Hot, Rotationally Cold H_2 in the Outer Disk of Oph 163131 Nonthermally Excited by Ultraviolet and Cosmic Rays**. **Assani, K. D.**, Li, Z.-Y., Bergner, J. B., Neufeld, D. A., Harsono, D., Drozdovskaya, M. N., Padovani, M., et al. *The Astronomical Journal*, 172, 170. DOI: 10.3847/1538-3881/ae947a

2025: **Mid-infrared extinction curve for protostellar envelopes from JWST-detected embedded jet emission: The case of TMC1A**. **Assani, K. D.**, Li, Z.-Y., Ramsey, J. P., Tychoniec, L., Francis, L., Le Gouellec, V. J. M., et al. *Astronomy & Astrophysics*, 701, A175. DOI: 10.1051/0004-6361/202555016

2024: **The asymmetric bipolar [Fe II] jet and H_2 outflow of TMC1A resolved with the JWST NIRSpec Integral Field Unit**. **Assani, K. D.**, Harsono, D., Ramsey, J. P., Li, Z.-Y., Bjerkeli, P., Pontoppidan, K. M., et al. *Astronomy & Astrophysics*, 688, A26. DOI: 10.1051/0004-6361/202449745

Peer-Reviewed Publications – Co-Author

2026: **Gas Flows and Mass Accretion Rates in Eight Dippers: HD 142666, HD 143006, HD 145718, V935 Sco, DoAr 25, EPIC 204638512, EPIC 205151387, and EPIC 203850058**. Sitko, M. L., Russell, R. W., **Assani, K. D.**, Villanueva, L., Hernandez, J. *The Astronomical Journal*. DOI: 10.3847/1538-3881/ae939f

2026: **JOYS+: A JWST/MIRI survey of the evolution of H_2 winds and jets from low-mass protostars**. Francis, L., Tychoniec, L., van Dishoeck, E. F., Sellek, A. D., Caratti o Garatti, A., Le Gouellec, V. J. M., et al., including **Assani, K. D.** *Astronomy & Astrophysics*, 711, A258. DOI: 10.1051/0004-6361/202558772

2026: **JOYS: Launch and destruction of dust in protostellar jets: The case of BHR71-IRS1 with JWST MIRI**. Tychoniec, L., Francis, L., Navarro, M. G., Vorster, J. M., van Dishoeck, E. F., Caratti o Garatti, A., **Assani, K. D.**, et al. *Astronomy & Astrophysics*, 710, A370. DOI: 10.1051/0004-6361/202558711

2026: **JWST Edge-on Disk Ice (JEDIce): Program Overview and Ice Survey Results**. Bergner, J. B., Arulanantham, N., Dartois, E., Drozdovskaya, M. N., Harsono, D., McClure, M., et al., including **Assani, K. D.** *The Astrophysical Journal*, 1002, 17. DOI: 10.3847/1538-4357/ae5961

2026: **The near-infrared silhouette of the Flying Saucer edge-on disc revealed by the JWST JEDIce program**. Dartois, E., Noble, J. A., Bergner, J. B., Pontoppidan, K. M., **Assani, K. D.**, Harsono, D., McClure, M. K., et al. *Astronomy & Astrophysics*. DOI: 10.1051/0004-6361/202557177

2024: **Gas Dynamics in 3 "Dippers": EPIC 203850058, EPIC 204638512, and EPIC 205151387 in 2017–2018**. Sitko, M. L., Russell, R. W., **Assani, K. D.**, Bayyari, A., Tyler, D., Lisse, C. M., Grady, C. A., et al. *Research Notes of the AAS*, 8, 310. DOI: 10.3847/2515-5172/ad9f33

2024: **Dracula's Chivito: Discovery of a Large Edge-on Protoplanetary Disk with Pan-STARRS**. Berghea, C. T., Bayyari, A., Sitko, M. L., Drake, J. J., Mosquera, A., Garraffo, C., Petit, T., Russell, R. W., **Assani, K. D.** *The Astrophysical Journal Letters*, 967, L3. DOI: 10.3847/2041-8213/ad43e3

2023: **Direct Images and Spectroscopy of a Giant Protoplanet Driving Spiral Arms in MWC 758**. Wagner, K., Stone, J., Skemer, A., Ertel, S., Dong, R., Apai, D., et al., including **Assani, K. D.** *Nature Astronomy*, 7, 1208–1217. DOI: 10.1038/s41550-023-02028-3

2023: **Wavelength-dependent Extinction and Grain Sizes in "Dippers"**. Sitko, M. L., Russell, R. W., Long, Z. C., **Assani, K. D.**, Pikhartova, M., Bayyari, A., et al. *The Astronomical Journal*, 166, 24. DOI: 10.3847/1538-3881/acd7e8

2021: **Variability of Disk Emission in Pre-main Sequence and Related Stars. V. Occultation Events from the Innermost Disk Region of the Herbig Ae Star HD 163296**. Pikhartova, M., Long, Z. C., **Assani, K. D.**, Fernandes, R. B., Bayyari, A., Sitko, M. L., et al. *The Astrophysical Journal*, 919, 64. DOI: 10.3847/1538-4357/ac03af

2018: **Differences in the Gas and Dust Distribution in the Transitional Disk of a Sun-like Young**

Star, PDS 70. Long, Z. C., Akiyama, E., Sitko, M., Fernandes, R. B., **Assani, K. D.**, Grady, C. A., et al. *The Astrophysical Journal*, 858, 112. DOI: 10.3847/1538-4357/aaba7c

Doctoral Dissertation

2026: **Exploring the Structure and Emission of Protostellar Systems with JWST**. **Assani, K. D.** Ph.D. Dissertation, University of Virginia. Defended April 17, 2026. DOI: 10.18130/hm8j-br11

White Papers

ALMA2040 White Paper Series

2026: **Chemical Demographics of Star-forming Regions**. Drozdovskaya, M. N., Csengeri, T., Beltrán, M. T., Jørgensen, J., Baudry, A., Beuther, H., et al., including **Assani, K. D.** *ALMA2040 White Paper Series*. arXiv:2610.22303

2026: **The environments forming stellar and planetary systems**. Kuffmeier, M., Pineda, J. E., Schilke, P., Jørgensen, J. K., **Assani, K. D.**, Baudry, A., et al. *ALMA2040 White Paper Series*. arXiv:2610.22462

2026: **Frontiers of chemical complexity in the interstellar medium: unveiling the cosmic origins of life**. Rivilla, V. M., Sanz-Novo, M., Beltrán, M., Jørgensen, J., Colzi, L., Baudry, A., et al., including **Assani, K. D.** *ALMA2040 White Paper Series*. arXiv:2610.21849

2026: **Multi-wavelength Polarization as a Probe of Planet Formation**. Stephens, I., Sadavoy, S., Kataoka, A., Lin, Z.-Y., Lankhaar, B., Guidi, G., et al., including **Assani, K. D.** *ALMA2040 White Paper Series*. arXiv:2610.22048

2026: **Planetary system architectures in the final stages of planet formation**. Luppe, P., Roumeliotis, Z., Marino, S., Kral, Q., Lovell, J. B., Matrà, L., et al., including **Assani, K. D.** *ALMA2040 White Paper Series*. arXiv:2610.21957